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Water and Society

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Climate Proofing through Sustainable Water Resource Management

What is climate proofing?

- Climate proofing is a term that refers to a process of mainstreaming climate change into mitigation and/or adaptation strategies and programmes.
- **In nutshell, climate proofing** is the process of assessing and integrating climate change risks into the planning, design, and implementation of policies, projects, and infrastructure, with the aim of reducing vulnerability, enhancing resilience, and ensuring long-term sustainability under changing climatic conditions.
- The term is often used in the development context. Here, climate proofing specifically refers to a process of mainstreaming climate change into development strategies and programmes, i.e. development is viewed through a climate change lens.

Watch these:

- https://youtu.be/w9zqR3zDA1s?feature=shared
- https://youtu.be/1lyXSmmXnm8?feature=shared
- https://youtu.be/dWn-ipdYzZA?feature=shared
- https://youtu.be/qPuKWDm9-Fg?feature=shared
- https://youtu.be/KY_k4VKY8NI?feature=shared
- https://youtu.be/nf-Yy3EuZi0?feature=shared

International and National Case Studies

- Integrated Urban Water Management Strategies through Regulatory Reforms in Australia:
- Many national, state and local governments are taking proactive measures through regulatory reforms that encourage efficient water use and help cities become water secure in the future.
- State governments in Australia are attempting to drought proof their cities by adopting strategies like grey water recycling, water efficient labelling on appliances, providing incentives to households conserving water, and putting water restrictions in place that are enforced by fines.
- The city of Perth is avoiding the situation of water crisis by adopting multiple strategies which help in water conservation, developing alternate sources of water and ways to recycle stormwater and wastewater.

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- Perth's Water Corporation has strongly engaged with the users in sensitizing them about their current water usage and options to reduce their consumption.
- They also engage with plumbers and landscapers to encourage homeowners to adopt water efficient fixtures.
- Perth's long-term strategy for sourcing water and managing its use is called 'Fresh Water Thinking'. It includes direct outreach and education to the users on aspects of their current water usage and potential avenues for saving water.
- Additionally, the government is operating two large scale desalination plants which cater to half of Perth's water supply (Core, L. N., 2020).
- Perth came up with its first action plan which was developed based on the ideas and actions suggested by over 200 stakeholders from diverse sectors including Government; communities; finance; local government, sport and cultural industries; planning, lands and heritage; the water corporation.

Effective Water Demand Management Plan for Cape Town

- The Government in South Africa implemented a robust water demand management plan for Cape Town.
- With water-saving efforts of citizens, the city's water demand reduced to 550 million liters per day (in 2018) from 900 million liters per day before the drought.
- Effective governance and holistic management of the integrated water cycle have facilitated Cape Town transition to a water-sensitive city (Cape Town Water Strategy, 2019).
- The city government also ensures access to potable water through delivery by both centralized and decentralized infrastructure.
- In addition to this, to improve the existing urban water cycle, the city has undertaken steps to integrate stormwater treatment with the landscape.
- The city has been effective in managing its water resources since 2018 water crisis and is continuing its efforts in effective network management.
- Some of the key activities undertaken include proactive leak detection, pipe and meter replacement, comprehensive asset management system, and smart pressure controller.

Nature-based Solutions (NbS) for Increased Climate Resilience in China

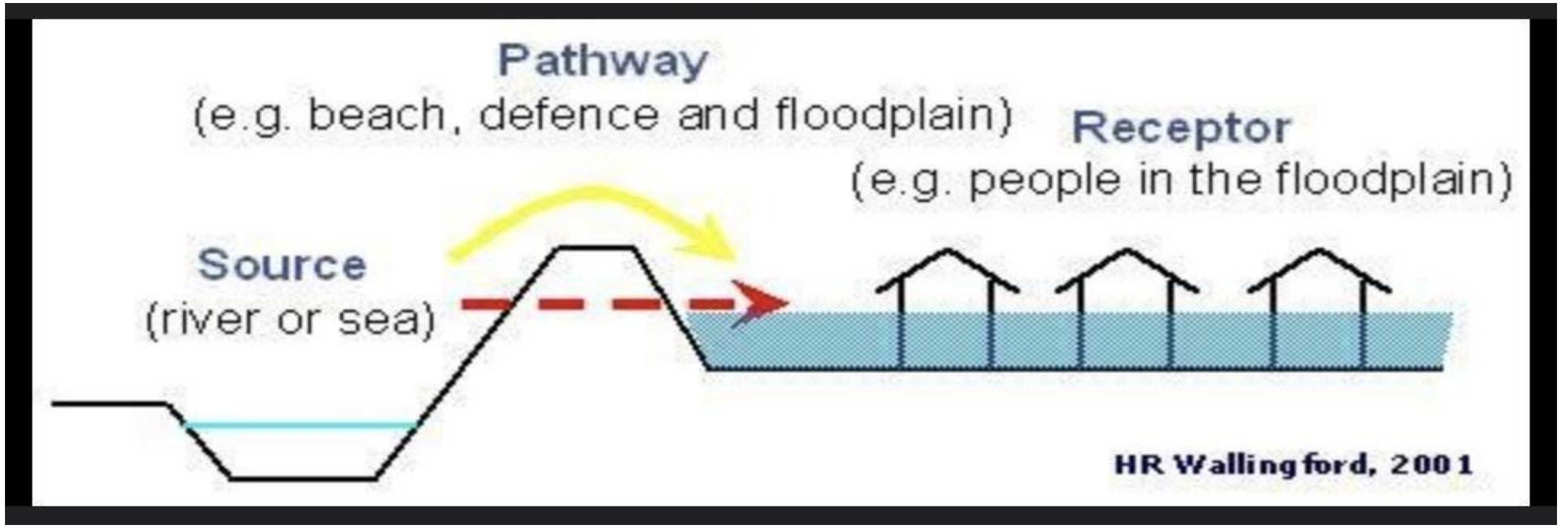
- Most cities in China do not have a drainage system that is at par with its national flood prevention safety standards. This led to urban flooding and contamination of the water supply in several areas.
- In response to its urban water management challenges, the Government of China launched the "Sponge City Programme" in 2013.
- Wuhan is leading the way for the exemplary "nature-based" approach implemented in the city.
- NbS helps in protecting, restoring and managing the ecosystems, which can be easily applied in cities transforming them to an urban system that is regenerative and accessible.
- The city has green infrastructure such as rain gardens, absorptive roads, permeable pavements, rainwater reuse facilities, etc.
- The city's nature-based approach has cost around USD 600 million (Oates et al., 2020) which is still low-cost compared to the alternative grey infrastructure-based approach (like upgrading the city's drainage system) to improve the city's resilience to flooding.

Active, Beautiful, Clean (ABC) Waters Program of Singapore

- Singapore's long-term vision, adoption of innovative technologies and stakeholder engagement efforts has led Singapore to become one of the most talked about example in water sector.
- Singapore's National Water Agency, Public Utility Board (PUB) has taken many worth mentioning initiatives.
- One such initiative is the Active, Beautiful, Clean (ABC) Waters program (PUB, 2014), which demonstrated how Singapore has incorporated sustainable city planning with stormwater management.
- The program was launched in 2006 and ensured that there is inter-connection of green and blue spaces. The goal was to create community spaces around reservoirs and canals, giving these assets a recreational function.

- Further, to build resilience to the current flood protection scheme, PUB, has adopted a "Source-Pathway-Receptor" strategy, which seeks to develop catchment-wide solutions.

- This is a holistic approach where flexibility and adaptability to the entire system is ensured, addressing not just the stormwater drains and canals (the pathways for water) but also areas that generate stormwater runoff (source) and where floods may occur (receptor).



Converting Wastewater as a Resource in Israel

- Mediterranean water crisis is known to everyone and so is the remarkable achievement of Israel (OECD, n.d.) in the water sector.
- Israel holds a world record of 85% reclaimed effluents reused in agriculture and is a global champion in the development and the production of efficient water saving irrigation systems.
- The country uses around 21% of treated wastewater in total water consumed by the industries and around 45% in agriculture sector. Some of the key steps undertaken by the Government included: • In 2010, new and stringent quality standards for effluents were legislated with 37 different quality parameters.

The standards enable to use the effluents without any limits for all purposes and thus the Ministry of Health allocates permits for unlimited cultivation of crops. Secondly, Water and Sewerage Corporations Law in 2001 transferred water and sewerage services from the municipalities to corporate entities.

This was the first step in the transformation of the administratively managed water sector to a more commercially oriented one. It helped to bring finance for infrastructure investments, and the assurance of high-quality services. Use of both economics (closed market) and environmental (stringent standards) principles helped in development of an advanced wastewater reuse system in Israel.

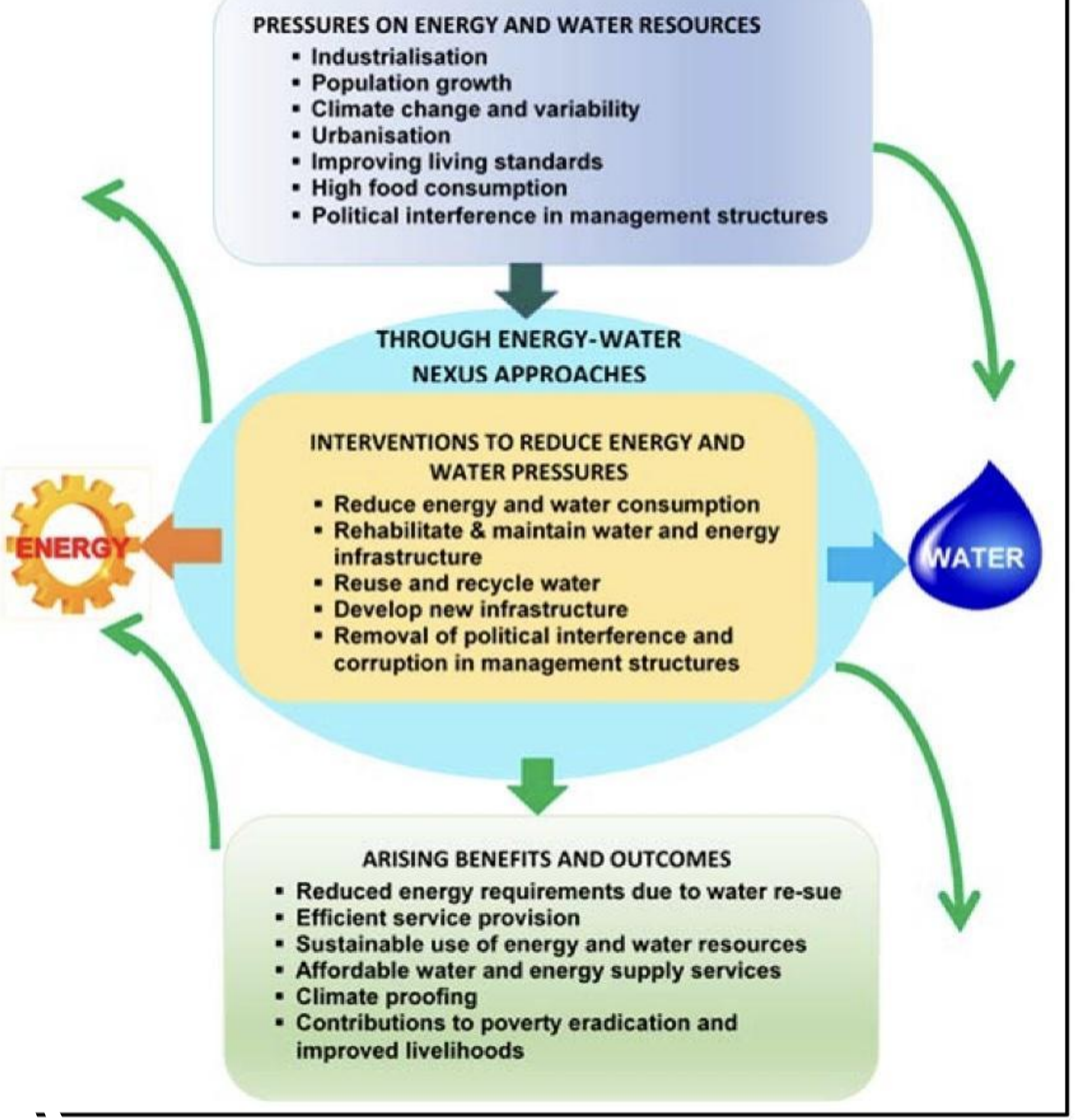
Women as Champions of Pond Rejuvenation in Rajasthan, India

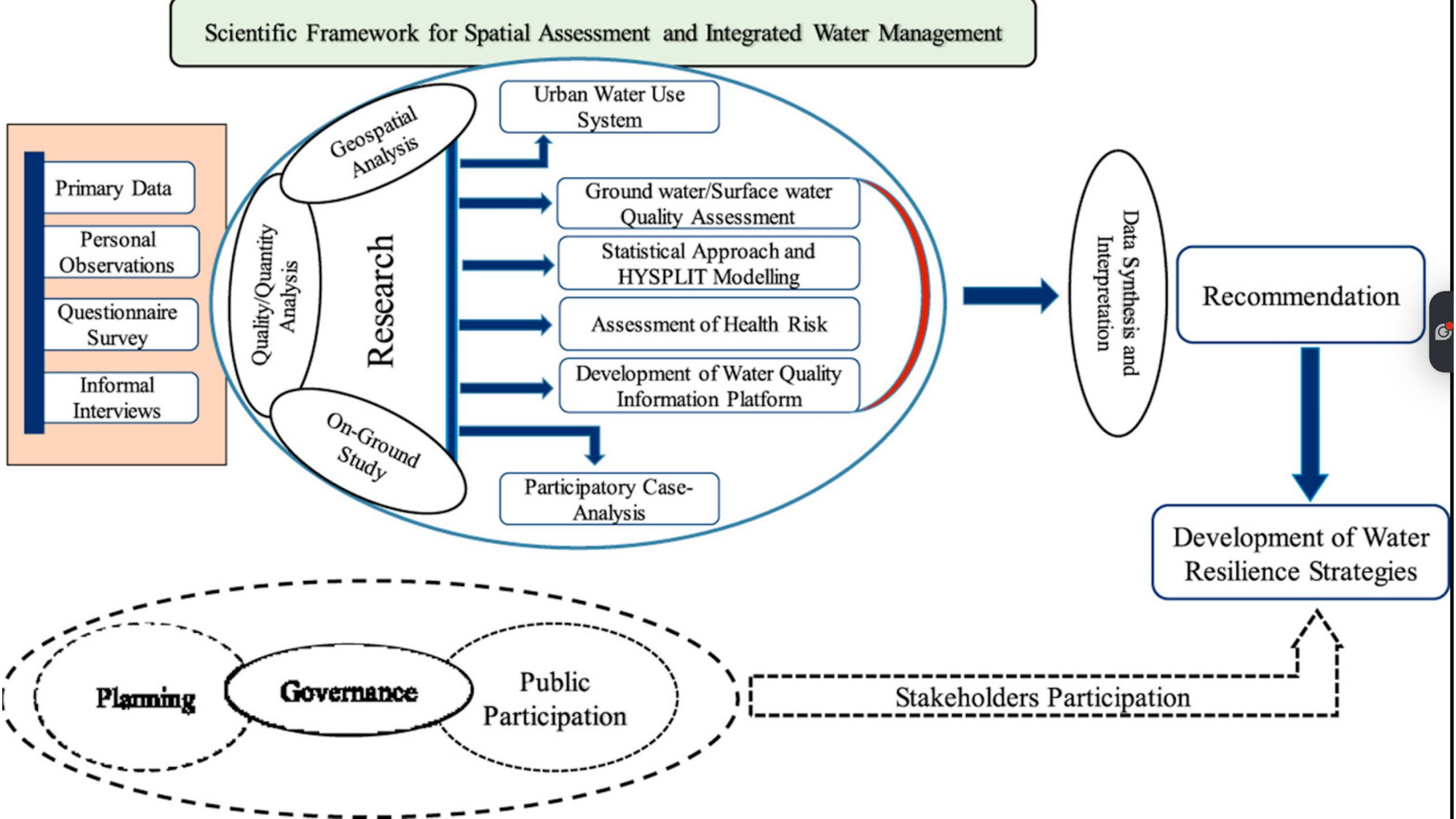
- The Bal Rashmi Society in Rajasthan, India has taken up rehabilitation of ponds work in parched and rural areas of Rajasthan that have not been desilted for 10 years (Jayalakshmi D., 2003).
- Earlier, women were facing the plight of water stress where they had to walk 5 to 6 km for fetching drinking water. Women contributed 90 % of the labour in the rehabilitation work.
- Mahila mandals were entrusted with the maintenance of the ponds by way of shramdhan. This led to a significant impact where in 56 villages were benefited; 67 ponds deepened; storage capacity has risen by 5-7 ft; more drinking water has been provided for cattle; run-off water has been stored; green belt and cooler surroundings were developed. This also is an adaptation strategy towards climate change.
- Women can play a pivotal role like this in enhancing water security of the region through their involvement and management which can help community stride over water stress situation and combat deleterious impacts of climate change. it also resonates with the Government flagship program ‘Jal Jeevan Mission’ where the focus is on participation and empowerment of women for safe water access programs.

River Cities Alliance in India

- The River Cities Alliance (RCA) is a joint initiative of the Department of Water Resources, River Development & Ganga Rejuvenation under the Ministry of Jal Shakti (MoJS) & the Ministry of Housing and Urban Affairs (MoHUA), with a vision to connect river cities and focus on sustainable river centric development.
- It has 110 member cities including one international member city from Denmark. RCA is envisaged as a facilitatory platform for initiating river-sensitive planning and development. The platform would allow cities to discuss, learn from each other the best practices (PIB, 2022).

Adaptation Strategies





Scientific Framework for Spatial Assessment and Integrated Water Management
Primary Data
Personal Observations
Questionnaire Survey
Informal Interviews
Geospatial Analysis
Quality/Quantity Analysis
On-Ground Study
Research
Urban Water Use System
Ground water/Surface water Quality Assessment
Statistical Approach and HYSPLIT Modelling
Assessment of Health Risk
Development of Water Quality Information Platform
Participatory Case-Analysis
Data Synthesis and Interpretation
Recommendation
Development of Water Resilience Strategies
Planning
Governance
Public Participation
Stakeholders Participation